

Programming Concepts -- Guided Notes

Unit tp-3.9 | CompTIA Network+ N10-009 Obj. FC0-U71 3.9 | N10-008 FC0-U71 3.9

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A _____ stores a single value that can change during program execution. A _____ stores a value that does not change. A _____ is a named location that holds one value at a time. A _____ (like a list/array) can hold multiple values under one name.

Question 2

Data types define what kind of data a variable holds. Common types: _____ (whole numbers), _____ (decimal numbers), _____ (text), _____ (True/False). Assigning a value of the wrong type to a variable can cause a _____ error.

Question 3

OOP (Object-Oriented Programming) models software around _____ that combine data (_____) and behavior (_____). A _____ is a blueprint; an _____ is a specific object created from that blueprint. OOP supports _____ (hiding internal details) and _____ (child classes inherit from parent).

Question 4

A syntax error is a violation of the _____ rules of a programming language and is caught at _____ time. A logic error is a bug where the code runs without crashing but produces the _____ output. A runtime error occurs while the program _____ and causes it to _____ unexpectedly.

Question 5

Version control (like Git) tracks _____ to source code over time. A _____ is a saved version of the codebase. _____ allows developers to create an isolated copy of the code to work on a feature without affecting the main codebase. _____ combines changes from two branches.

Question 6

An API (Application Programming Interface) defines how two software components can _____ with each other. When a weather app on your phone displays today's forecast, it is likely calling a weather service _____ to get the data. APIs use _____ requests over HTTP to send and receive _____ (typically JSON or XML format) data.

Question 7

Testing ensures software functions as expected. _____ testing checks individual functions in isolation. _____ testing checks how multiple components work together. _____ testing verifies the complete system meets requirements from the user's perspective. _____ testing ensures that new changes do not break existing functionality.

Question 8 -- Real World Application

REAL WORLD: A student writes a program to calculate the area of a rectangle. They test it and it runs without errors, but the area value it prints is always wrong. Identify what type of error this is, explain why it does not show up as a crash, and describe how the student should debug it.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-3-9-programming-concepts/tp-3-9-programming-concepts-answer-key.pdf
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